# 1 Explanation certificate: churn100-s7-c1

This section is the explanation certificate of the churn100-s7-c1 network. It is rendered from the canonical JSON record `churn100-s7-c1.json` (schema 1.4.0, SHA-256 `55f21bd4a4575c9c...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

| | |
|---|---|
| Dataset | churn_onehot |
| Experiment | churn100-s7, seed 7 |
| Genome | `32d24e98-266e-450c-88c8-5bdc1abbf5dd` (NEAT key 20708, generation 99) |
| Explanation | `b370983e-3d7d-40c8-8b8f-f37c41a96cc1` (85 operations) |
| Train AUC / accuracy | 0.9189 / 0.921 (n = 4000) |
| Test AUC / accuracy | 0.9007 / 0.921 (n = 1000) |
| Output activation | sigmoid |
| Phenotype | 86 nodes (71 inputs, 14 hidden, 1 output), 59 enabled edges |
| Explained model state | 95 nodes, 61 enabled edges |
| Refactoring operations | add_identity_node 2, split_node 5 |
| Function preservation | max $\|\Delta \text{output}\| = 1.22 \times 10^{-15}$ on the training split |
| Annotations | 7 (5 leaf, 2 composition) |
| Structural coverage | 1 (94 of 94 candidate nodes; 35 Null) |
| Compositional coverage | 1 |
| Uncovered nodes / edges | 0 / 0 |
| Missing compositions | 0 |
| Full coverage | yes |
| Drafted by | AI-assisted draft |
| Verified by | unverified |
| Record hash (SHA-256) | `55f21bd4a4575c9c...` |

Table 1: Explanation certificate summary: churn100-s7-c1.

**Drafter's summary.** logit P(churn) = -1.96 + a small linear backbone over 27 inputs (all |w| <= 0.32 per SD; state offsets up to -1.9 logit) + an international-plan switch (+2.91 logit) + an always-on voicemail-plan/state:41 offset unit (-0.54, -1.04) + exact single-state offsets + usage_response, all additive in the logit. All nonlinearity is in usage_response: a usage hinge (day/evening/night minutes, never active for voicemail-plan customers, on from about 190 day minutes), a steeper second hinge for heavy users (from about 240 day minutes), and a service-calls gate (shut for 0-2 calls, opening from 3) that is closed again by usage, so service calls matter only for light users. Train AUC 0.919, accuracy 0.921. This retrained elite copy of the 10-seed churn100-s7 genome (same key 20708, generation 98) has the same three-part usage response and intl-plan switch; the weights differ slightly (e.g. intl plan +2.91 here vs +2.88).

**What resisted explanation.** Nothing structural (5 input splits, no hidden splits, no consolidation). The service-calls gate (4344) and the heavy-use hinge (14351 -> 10552) take the usage hinge 879 as their only hidden input, so they sit as own nodes of the usage_response composition rather than in leaves (a leaf cannot take another annotation's exit as an entry). Calibration gaps: the gate opens at 3 calls (+0.80 logit; model 0.150 vs empirical 0.106 churn, 530 customers), under-predicts at 4 calls (0.32 vs 0.44) and is over-confident from 6 calls (0.85-0.94 vs 0.55-0.68; 39 customers). State offsets are overstated for state:46 (-1.89 vs empirical -0.91), state:5 (-1.09 vs -0.51), state:10 (-0.63 vs -0.22) and state:49 (-0.69 vs -0.32). 5667 (state:2) is a dead unit (never active). closed_form_max_abs_err 0.038 (state_offsets, 13164 pre-activation up to 7.7) is 3-significant-figure rounding; numeric error 8.9e-16.

## 1.1 Plain-language summary

*AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.*

**Summary (drafted).** For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up five parts: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment and state adjustments. [C047] This part combines a usage score, built from daytime, evening and night-time minutes and the voice-mail plan, with calls to customer service, not as a simple sum. [C060] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan they add nothing until a point, then steadily: going from 0 to 4 calls takes a typical customer from about 1% to about 15%. [C055] Through it, daytime minutes add to the chance only above about 187 minutes for customers without a voice-mail plan, in an uneven way: for a typical customer, going from 0 to 271 minutes takes it from about 1% to about 11%. [C056] Daytime minutes count more with more than 201 evening minutes, in an uneven way: from about 1% to about 11% between 112 and 249 minutes, against about 1% to about 2% with fewer. [C067] This part reads the international plan alone and acts on the 380 of 4000 training customers who have one, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance of leaving from about 1% to about 15%. [C077] This part adds up many inputs with fixed weights, each moving a typical customer's chance by a few points at most: compared with most other states, Washington raises it to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] A higher international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and more voice-mail messages and a longer account also raise it steadily, by less than a point. [C083, C086, C098] More international calls lower it, each extra call taking away about the same, from about 1% at 2 to under 1% at 8 calls, and more evening and night-time minutes lower it steadily through this part, by less than a point. [C087, C092, C096] This part reads the voice-mail plan and the state: through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also lowers a typical customer's chance to under 1%. [C108, C109] This part reads only the state: it moves a typical customer's chance from about 1% in West Virginia to about 5% in South Dakota and about 2% in Indiana and in most other states, and keeps it under 1% in Virginia (66 to 122 training customers per named state; the full list is in the technical detail). [C111, C112, C114, C115]

**The whole model.** The model adds up five parts into one score: usage and service calls, the international plan, fixed-weight inputs, a voice-mail and South Dakota adjustment, and state adjustments. [C047] *Evidence.* It is made up of these five parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C047] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 15%, and more daytime minutes raise it in an uneven way, from about 1% at 112 to about 4% at 249 minutes. [C001, C004] In Washington the chance is about 5% and in Montana about 4%, against about 1% in West Virginia and under 1% in Vermont. [C002, C003, C022] The model ignores daytime calls, evening calls, the evening charge, international minutes and the night-time charge. [C042, C043, C044, C045, C046]

**Usage and service calls.** This part raises the chance of leaving for heavy daytime users without a voice-mail plan and for customers who call customer service 3 times or more: daytime minutes in an uneven way, and calls for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily. [C056, C055] *Evidence.* It is made up of

the usage score (explained below) together with customer-service calls, combined in a way that is not a simple sum. [C060] Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes for a typical customer. [C056] Customer-service calls add only from about 3 calls upwards, for most customers slowly at first, then faster, but for some customers without a voice-mail plan nothing until a point, then steadily, from about 1% to about 15% between 0 and 4 calls. [C055]

**Usage score.** This part builds a usage score: heavy daytime use raises the chance of leaving, in an uneven way and more so with high evening minutes, and a voice-mail plan lowers it. [C071, C067, C062] *Evidence.* Through it, daytime minutes add only above about 187 minutes for customers without a voice-mail plan, in an uneven way, from about 1% to about 11% between 0 and 271 minutes. [C071] Daytime minutes count more with more than 201 evening minutes (about 1% to about 11%, against about 1% to about 2%) and with more than 200 night-time minutes (about 1% to about 7%, against about 1% to about 3%), in an uneven way. [C067, C068] A voice-mail plan lowers the chance through it from about 3% to about 1% for the typical customer it acts on. [C062] Its formula reproduces the part exactly on all 4000 training customers. [C076]

**International plan.** This part raises the chance of leaving for customers with an international plan. [C077] *Evidence.* It acts on the 380 of 4000 training customers with an international plan, 161 of whom left; for the typical customer it acts on, the chance is about 2% without it and about 26% with it. [C078] For a typical customer, an international plan takes the chance from about 1% to about 15%. [C077] Its formula reproduces the part exactly on all 4000 training customers. [C079]

**Fixed-weight inputs.** This part adds small fixed-weight effects: a higher international charge, more voice-mail messages and a longer account raise the chance steadily, more international calls lower it, each extra call taking away about the same, and some states shift it up or down. [C083, C086, C098, C087, C080, C082] *Evidence.* Through it, compared with most other states, Washington raises a typical customer's chance to about 3% and Montana to about 2%, and Vermont, Rhode Island and Colorado lower it to under 1%. [C080, C081, C082, C084, C085] The international charge raises it steadily, from under 1% at 1.84 to about 1% at 3.70 US dollars, and international calls lower it from about 1% at 2 to under 1% at 8 calls, each extra call taking away about the same. [C083, C087] More evening and night-time minutes lower it steadily through this part, by less than a point. [C092, C096] Its formula reproduces the part exactly on all 4000 training customers. [C107]

**Voice mail and South Dakota.** This part lowers the chance of leaving for customers in South Dakota and for customers with a voice-mail plan. [C108, C109] *Evidence.* Through it, a customer in South Dakota has under 1% instead of about 1%, and a voice-mail plan also takes a typical customer to under 1%. [C108, C109] Its formula reproduces the part exactly on all 4000 training customers. [C110]

**State adjustments.** This part shifts the chance of leaving by state: it is lowest in West Virginia and Virginia and highest in South Dakota. [C111, C114, C115] *Evidence.* Through it, a typical customer's chance is about 5% in South Dakota and about 2% in Indiana and in most other states, against about 1% in West Virginia and under 1% in Virginia. [C111, C112, C114, C115] Its formula reproduces the part exactly on all 4000 training customers. [C117]

**Worth checking against what is known.** Worth checking: the state moves the chance by several points (for example Washington about 5% and Vermont under 1% across the whole model), based on 72 to 122 training customers per state. [C002, C022] Worth checking: South Dakota gets a rise through the state adjustments (to about 5%) and a fall through the voice-mail and South Dakota adjustment (to under 1%), so the two pull against each other. [C111, C108] Worth checking: more evening and night-time minutes raise the chance through the usage score but lower it through

the fixed-weight inputs. [C063, C064, C092, C096] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C027, C023]

## Claims.

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C001 | direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan* | 1% to 15% | log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91 | 3620, 380 | train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000) | yes | clearly |
| C002 | direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 5% | log-odds 1.79, OR 5.98; range across patients 1.79 to 1.79 | 122, 72 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C003 | direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 4% | log-odds 1.50, OR 4.47; range across patients 1.50 to 1.50 | 122, 75 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C004 | direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%* | 1% to 4% | log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32 | 4000 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | slightly |
| C005 | direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%* | 1% to 3% | log-odds 0.86, OR 2.37; range across patients 0.00 to 1.25 | 4000 | train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000) | yes | slightly |
| C006 | direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 3% | log-odds 1.12, OR 3.06; range across patients 1.12 to 1.12 | 122, 66 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C007 | direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 3% | log-odds 0.94, OR 2.55; range across patients 0.79 to 1.12 | 122, 66 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C008 | direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 3% | log-odds 1.02, OR 2.78; range across patients 1.02 to 1.02 | 122, 90 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C009 | direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 2% | log-odds 0.98, OR 2.66; range across patients 0.98 to 0.98 | 122, 87 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C010 | direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 2% | log-odds 0.97, OR 2.64; range across patients 0.97 to 0.97 | 122, 74 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C011 | direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 2% | log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96 | 122, 84 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C012 | direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 2% | log-odds 0.91, OR 2.47; range across patients 0.91 to 0.91 | 122, 81 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C013 | direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.83, OR 2.30; range across patients 0.83 to 0.83 | 122, 82 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C014 | direction: state: West Virginia to Mississippi (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.80, OR 2.22; range across patients 0.80 to 0.80 | 122, 77 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C015 | direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.70, OR 2.02; range across patients 0.70 to 0.70 | 122, 82 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C016 | direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: dip at 212 minutes; 36% of patients same; fallback; risk saturates near 0%* | 1% to 2% | log-odds 0.67, OR 1.96; range across patients -0.35 to 3.65 | 4000 | train; test agrees (AUC gain 0.030 [0.010, 0.050], n=1000) | no | slightly |
| C017 | direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69 | 122, 78 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C018 | direction: state: West Virginia to state other than Alaska, Georgia, Indiana, Kentucky, Arkansas, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, West Virginia, Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69 | 122, 2191 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C019 | direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.80, OR 2.23; range across patients 0.80 to 0.80 | 4000 | train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000) | yes | slightly; on the training data only |
| C020 | direction: state: West Virginia to Wisconsin (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.58, OR 1.78; range across patients 0.58 to 0.58 | 122, 86 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C021 | direction: state: West Virginia to New Mexico (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.55, OR 1.74; range across patients 0.55 to 0.55 | 122, 72 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C022 | direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to <1% | log-odds -1.20, OR 0.30; range across patients -1.20 to -1.20 | 122, 80 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C023 | direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47 | 4000 | train; test unclear (AUC gain 0.0008 [-0.004, 0.006], n=1000) | yes | slightly; on the training data only |
| C024 | direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47 | 122, 68 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C025 | direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63 | 4000 | train; test unclear (AUC gain 0.002 [-0.004, 0.008], n=1000) | yes | slightly; on the training data only |
| C026 | direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54 | 122, 82 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C027 | direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan* | 1% to 1% | log-odds -1.37, OR 0.25; range across patients -3.46 to -0.54 | 2939, 1061 | train; test agrees (AUC gain 0.019 [0.005, 0.031], n=1000) | yes | slightly |
| C028 | direction: state: West Virginia to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40 | 122, 78 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C029 | direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 40.06 US dollars; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 38.81 US dollars* | 1% to 1% | log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11 | 4000 | train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000) | yes | slightly; on the training data only |
| C030 | direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds -0.28, OR 0.76; range across patients -0.28 to -0.28 | 122, 91 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C031 | direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23 | 4000 | train; test disagrees (AUC gain -0.001 [-0.003, 0.001], n=1000) | yes | slightly; on the training data only |
| C032 | direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: dip at 218 minutes; 30% of patients same; fallback; risk saturates near 0%* | 1% to 1% | log-odds 0.34, OR 1.40; range across patients -0.29 to 2.38 | 4000 | train; test agrees (AUC gain 0.007 [0.0006, 0.013], n=1000) | no | slightly |
| C033 | direction: area code: area code other than area code 510 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 510, area code other than area code 510* | 1% to 1% | log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10 | 3040, 960 | train; test unclear (AUC gain 8e-05 [-0.001, 0.002], n=1000) | yes | slightly; on the training data only |
| C034 | direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07 | 122, 60 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C035 | direction: total night-time calls: 74 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.06, OR 1.07; range across patients 0.06 to 0.06 | 4000 | train; test disagrees (AUC gain -9e-05 [-0.0008, 0.0006], n=1000) | yes | slightly; on the training data only |
| C036 | direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02 | 122, 56 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C037 | magnitude: international calling plan: no international plan to an international plan, typical patient<br>*shape: categorical; highest to lowest: an international plan, no international plan* | 1% to 15% | log-odds 2.91, OR 18.3 | 3620, 380 | train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000) | yes | clearly |
| C038 | magnitude: state: West Virginia to Washington, typical patient<br>*shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio ...* | 1% to 5% | log-odds 1.79, OR 5.98 | 122, 72 | train; test disagrees (AUC gain -0.004 [-0.017, 0.006], n=1000) | yes | slightly; on the training data only |
| C039 | magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient<br>*shape: accelerating; 67% of patients same; fallback; risk saturates near 0%* | 1% to 4% | log-odds 1.47, OR 4.37 | 4000 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | slightly |
| C040 | magnitude: customer-service calls: 0 calls to 3 calls, typical patient<br>*shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%* | 1% to 3% | log-odds 1.25, OR 3.50 | 4000 | train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000) | yes | slightly |
| C041 | magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient<br>*shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.80, OR 2.23 | 4000 | train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000) | yes | slightly; on the training data only |
| C042 | null: model ignores total daytime calls (exact rule) | | max change 0.000 (max \|change in log-odds\|) | 4000 | train | yes | |
| C043 | null: model ignores total evening calls (exact rule) | | max change 0.000 (max \|change in log-odds\|) | 4000 | train | yes | |
| C044 | null: model ignores total evening charge (exact rule) | | max change 0.000 (max \|change in log-odds\|) | 4000 | train | yes | |
| C045 | null: model ignores total international minutes (exact rule) | | max change 0.000 (max \|change in log-odds\|) | 4000 | train | yes | |
| C046 | null: model ignores total night-time charge (exact rule) | | max change 0.000 (max \|change in log-odds\|) | 4000 | train | yes | |
| C047 | structure: made of linear_backbone, usage_response, intl_plan_switch, vmail_state41_unit, state_offsets; added | | log-odds n/a, OR n/a | 4000 | train | yes | |
| C048 | direction: total daytime minutes: 112 minutes to 249 minutes (through this part)<br>*shape: accelerating; 67% of patients same; fallback; risk saturates near 0%* | 1% to 4% | log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32 | 4000 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | slightly |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C049 | direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 89% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%* | 1% to 3% | log-odds 0.86, OR 2.37; range across patients 0.00 to 1.25 | 4000 | train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000) | yes | slightly |
| C050 | direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%* | 1% to 2% | log-odds 1.03, OR 2.79; range across patients 0.00 to 4.01 | 4000 | train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000) | yes | slightly |
| C051 | direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%* | 1% to 1% | log-odds 0.63, OR 1.87; range across patients 0.00 to 2.67 | 4000 | train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000) | yes | slightly |
| C052 | direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)* | 1% to 1% | log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11 | 4000 | train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000) | yes | slightly; on the training data only |
| C053 | direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota* | 1% to 1% | log-odds -0.18, OR 0.83; range across patients -0.32 to 0.00 | 3934, 66 | train; test unclear (AUC gain 0.0002 [-0.0002, 0.0007], n=1000) | yes | slightly; on the training data only |
| C054 | direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: a voice-mail plan, no voice-mail plan* | 1% to 1% | log-odds -0.83, OR 0.44; range across patients -2.92 to 0.00 | 2939, 1061 | train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000) | yes | slightly |
| C055 | threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 86% of patients same; mixed (but hinge for customers without a voice-mail plan); risk saturates near 0%* | 1% to 15% (typical patient with a fold) | fold 2.24 calls (IQR 2.24 to 2.96); past it log-odds 2.20, OR 9.02; range across patients 0.00 to 2.90 | 3833, 614 | train; test agrees (AUC gain 0.097 [0.061, 0.136], n=1000) | yes | clearly |
| C056 | threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%* | 1% to 11% (typical patient with a fold) | fold 187 minutes (IQR 158 to 214); past it log-odds 2.80, OR 16.4; range across patients 0.00 to 6.18 | 3014, 1382 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | clearly |
| C057 | threshold: total daytime charge: fold at about 38.81 US dollars (starts) for customers without a voice-mail plan *shape: hinge at 38.81 US dollars; 45% of patients same; fallback; risk saturates near 0%; one point with C057 (median fold 38.81 US dollars, typical patient 36.87 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)* | 1% to 2% (typical patient with a fold) | fold 38.8 US dollars (IQR 37.2 to 47.6); past it log-odds 0.73, OR 2.07; range across patients 0.00 to 2.28 | 2578, 579 | train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000) | yes | slightly; on the training data only |
| C058 | threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%* | 1% to 2% (typical patient with a fold) | fold 220 minutes (IQR 150 to 296); past it log-odds 0.75, OR 2.11; range across patients -0.06 to 2.45 | 2340, 1023 | train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000) | no | slightly |
| C059 | threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%* | 2% to 2% (typical patient with a fold (customers with a voice-mail plan)) | fold 318 minutes (IQR 294 to 327); past it log-odds -0.01, OR 0.99; range across patients 0.00 to 0.00 | 43, 0 (small n) | train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000) | no | slightly; the model has little data on this |
| C060 | structure: made of usage_hinge and inputs number customer service calls; combined non-additively | | log-odds n/a, OR n/a | 4000 | train | yes | |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C061 | direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 67% of patients same; fallback; risk saturates near 0%* | 1% to 4% | log-odds 1.37, OR 3.93; range across patients 0.00 to 4.32 | 4000 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | slightly |
| C062 | direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan* | 3% to 1% (typical patient this part acts on) | log-odds -0.83, OR 0.44; range across patients -2.92 to 0.00 | 2939, 1061 | train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000) | yes | slightly |
| C063 | direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: accelerating; 40% of patients same; fallback; risk saturates near 0%* | 1% to 2% | log-odds 1.03, OR 2.79; range across patients 0.00 to 4.01 | 4000 | train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000) | yes | slightly |
| C064 | direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: hinge at 224 minutes; 35% of patients same; fallback; risk saturates near 0%* | 1% to 1% | log-odds 0.63, OR 1.87; range across patients 0.00 to 2.67 | 4000 | train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000) | yes | slightly |
| C065 | direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 38.81 US dollars; 50% of patients same; fallback; risk saturates near 0%; one point with C072 (median fold 38.81 US dollars, typical patient 40.06 US dollars, 10th-90th percentile 37.22 US dollars to 53.94 US dollars, wide)* | 1% to 1% | log-odds 0.24, OR 1.27; range across patients 0.00 to 1.11 | 4000 | train; test unclear (AUC gain 0.0003 [-0.007, 0.008], n=1000) | yes | slightly; on the training data only |
| C066 | direction: state: state other than South Dakota to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, state other than South Dakota* | 3% to 3% (typical patient this part acts on) | log-odds -0.18, OR 0.83; range across patients -0.32 to 0.00 | 3934, 66 | train; test unclear (AUC gain 0.0002 [-0.0002, 0.0007], n=1000) | yes | slightly; on the training data only |
| C067 | interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening minutes: total evening minutes above 201 minutes minus total evening minutes at or below 201 minutes; stronger in total evening minutes above 201 minutes *shape: accelerating; 75% of patients same; fallback; risk saturates near 0%* | total evening minutes above 201 minutes: 1% to 11%; total evening minutes at or below 201 minutes: 1% to 2% | log-odds difference 1.54, OR 4.66 | 1998, 2002 | train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000); group difference on test 1.71 [1.48, 1.94] | yes | clearly |
| C068 | interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total night-time minutes: total night-time minutes above 200 minutes minus total night-time minutes at or below 200 minutes; stronger in total night-time minutes above 200 minutes *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%* | total night-time minutes above 200 minutes: 1% to 7%; total night-time minutes at or below 200 minutes: 1% to 3% | log-odds difference 1.00, OR 2.71 | 2000, 2000 | train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000); group difference on test 1.09 [0.85, 1.33] | yes | slightly |
| C069 | interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 70% of patients same; fallback; risk saturates near 0%* | a voice-mail plan: 1% to 1%; no voice-mail plan: 1% to 4% | log-odds difference -1.86, OR 0.16 | 1061, 2939 | train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000); group difference on test -1.89 [-2.04, -1.73] | no | slightly |
| C070 | interaction: effect of total evening minutes (138 minutes to 266 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: dip at 212 minutes; 48% of patients same; fallback; risk saturates near 0%* | a voice-mail plan: 1% to <1%; no voice-mail plan: 1% to 2% | log-odds difference -1.40, OR 0.25 | 1061, 2939 | train; test agrees (AUC gain 0.013 [0.0009, 0.023], n=1000); group difference on test -1.35 [-1.49, -1.21] | no | slightly |
| C071 | threshold: total daytime minutes: fold at about 187 minutes (starts) for customers without a voice-mail plan *shape: accelerating; 67% of* | 1% to 11% (typical patient with a fold) | fold 187 minutes (IQR 158 to 214); lift it log-odds 2.80, OR 16.4; range across patients 0.00 to 6.18 | 3014, 1382 | train; test agrees (AUC gain 0.035 [0.016, 0.052], n=1000) | yes | clearly |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C073 | threshold: total night-time minutes: fold at about 220 minutes (starts) for customers without a voice-mail plan *shape: hinge at 230 minutes; 45% of patients same; fallback; risk saturates near 0%* | 1% to 2% (typical patient with a fold) | fold 220 minutes (IQR 150 to 296); past it log-odds 0.75, OR 2.11; range across patients -0.06 to 2.45 | 2340, 1023 | train; test agrees (AUC gain 0.01 [0.003, 0.016], n=1000) | no | slightly |
| C074 | threshold: total evening minutes: fold at about 318 minutes (starts) for customers with a voice-mail plan *shape: step at 307 minutes; 50% of patients same; fallback; risk saturates near 0%* | 2% to 2% (typical patient with a fold (customers with a voice-mail plan)) | fold 318 minutes (IQR 294 to 327); past it log-odds -0.01, OR 0.99; range across patients 0.00 to 0.00 | 43, 0 (small n) | train; test agrees (AUC gain 0.034 [0.013, 0.055], n=1000) | no | slightly; the model has little data on this |
| C075 | subgroup: who this part acts on: 1382 of 4000 patients | 1% to 3% (typical patient this part acts on (risk without the part, then with it)) | share 0.35; log-odds 1.77, OR 5.86; range across patients -0.34 to 5.50 | 1382, 2618 | train; test agrees (AUC gain 0.214 [0.160, 0.273], n=1000) | no | slightly |
| C076 | structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16) | | log-odds 9e-16, OR n/a | 4000 | train | yes | |
| C077 | direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan* | 1% to 15% | log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91 | 3620, 380 | train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000) | yes | clearly |
| C078 | subgroup: who this part acts on: 380 of 4000 patients | 2% to 26% (typical patient this part acts on (risk without the part, then with it)) | share 0.10; log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91 | 380, 3620 | train; test agrees (AUC gain 0.060 [0.033, 0.090], n=1000) | yes | clearly |
| C079 | structure: its formula reproduces it on 4000 training patients (largest difference 0.0) | | log-odds 0.00, OR n/a | 4000 | train | yes | |
| C080 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 3% | log-odds 1.10, OR 2.99; range across patients 1.10 to 1.10 | 2529, 72 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C081 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 2% | log-odds 0.81, OR 2.24; range across patients 0.81 to 0.81 | 2529, 75 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C082 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to <1% | log-odds -1.89, OR 0.15; range across patients -1.89 to -1.89 | 2529, 80 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C083 | direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.80, OR 2.23; range across patients 0.80 to 0.80 | 4000 | train; test unclear (AUC gain 0.003 [-0.005, 0.011], n=1000) | yes | slightly; on the training data only |
| C084 | direction: state: state | 1% to <1% | log-odds -1.23, OR | 2529, | train; test disagrees | yes | slightly; on |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
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| C085 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to <1% | log-odds -1.09, OR 0.34; range across patients -1.09 to -1.09 | 2529, 78 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C086 | direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.47, OR 1.60; range across patients 0.47 to 0.47 | 4000 | train; test unclear (AUC gain 0.0008 [-0.004, 0.006], n=1000) | yes | slightly; on the training data only |
| C087 | direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63 | 4000 | train; test unclear (AUC gain 0.002 [-0.004, 0.008], n=1000) | yes | slightly; on the training data only |
| C088 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.43, OR 1.53; range across patients 0.43 to 0.43 | 2529, 66 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C089 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.43, OR 1.53; range across patients 0.43 to 0.43 | 2529, 66 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C090 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Georgia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to <1% | log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63 | 2529, 60 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C091 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33 | 2529, 90 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C092 | direction: total evening | 1% to 1% | log-odds -0.35, OR | 4000 | train; test disagrees | yes | slightly; on |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
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| C097 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.21, OR 1.24; range across patients 0.21 to 0.21 | 2529, 81 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C098 | direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%* | 1% to 1% | log-odds 0.23, OR 1.26; range across patients 0.23 to 0.23 | 4000 | train; test disagrees (AUC gain -0.001 [-0.003, 0.001], n=1000) | yes | slightly; on the training data only |
| C099 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds -0.22, OR 0.80; range across patients -0.22 to -0.22 | 2529, 68 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C100 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Virginia (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.15, OR 1.16; range across patients 0.15 to 0.15 | 2529, 91 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C101 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to New Mexico (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds -0.14, OR 0.87; range across patients -0.14 to -0.14 | 2529, 72 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C102 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, North Carolina, North Dakota, New Jersey, New Mexico, Ohio, Oregon, Rhode Island, South Carolina, South Dakota, Virginia, Vermont, Washington, Wisconsin, Colorado to Mississippi (through this part) *shape: categorical; highest to lowest: Washington, Montana, South Carolina, South Dakota, New Jersey, Ohio …* | 1% to 1% | log-odds 0.11, OR 1.11; range across patients 0.11 to 0.11 | 2529, 77 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], n=1000) | yes | slightly; on the training data only |
| C103 | direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 510, area code other than area code 510* | 1% to 1% | log-odds 0.10, OR 1.11; range across patients 0.10 to 0.10 | 3040, 960 | train; test unclear (AUC gain 8e-05 [-0.001, 0.002], n=1000) | yes | slightly; on the training data only |
| C104 | direction: state: state other than Georgia, Kentucky, Maryland, Mississippi, Montana, | 1% to 1% | log-odds -0.11, OR 0.89; range across patients -0.11 to -0.11 | 2529, 86 | train; test disagrees (AUC gain -0.004 [-0.016, 0.006], | yes | slightly; on the training data only |

| Claim | What is checked | Risk, typical patient | Estimate (log-odds; OR); range across patients | n | Test check | Verified | Words |
|---|---|---|---|---|---|---|---|
| C109 | direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan* | 1% to 1% | log-odds -0.54, OR 0.58; range across patients -0.54 to -0.54 | 2939, 1061 | train; test unclear (AUC gain 0.002 [-0.003, 0.008], n=1000) | yes | slightly; on the training data only |
| C110 | structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16) | | log-odds 4e-16, OR n/a | 4000 | train | yes | |
| C111 | direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 5% | log-odds 1.73, OR 5.62; range across patients 1.73 to 1.73 | 122, 66 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C112 | direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 2% | log-odds 0.83, OR 2.30; range across patients 0.83 to 0.83 | 122, 82 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C113 | direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 2% | log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69 | 122, 78 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C114 | direction: state: West Virginia to state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 2% | log-odds 0.69, OR 2.00; range across patients 0.69 to 0.69 | 122, 3505 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C115 | direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 1% | log-odds -0.43, OR 0.65; range across patients -0.43 to -0.43 | 122, 91 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C116 | direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Indiana, Arkansas, state other than Alaska, Indiana, Arkansas, South Dakota, Virginia, West Virginia, Alaska, West Virginia ...* | 1% to 1% | log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02 | 122, 56 | train; test disagrees (AUC gain -0.0004 [-0.006, 0.004], n=1000) | yes | slightly; on the training data only |
| C117 | structure: its formula reproduces it on 4000 training patients (largest difference 0.0) | | log-odds 0.00, OR n/a | 4000 | train | yes | |
| C118 | boundary: boundary | | log-odds n/a, OR n/a | 4000 | train | no | |
| C119 | boundary: boundary | | log-odds 1.00, OR n/a | 4000 | train | yes | |

## Not supported by the model.

- C016: direction claim about total evening minutes in the whole model: not supported by the model (58% of patients have the opposite sign (allowed: 10%)).
- C032: direction claim about total night-time minutes in the whole model: not supported by the model (63% of patients have the opposite sign (allowed: 10%)).

- C058: threshold claim about total night-time minutes in part ann_83: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).
- C059: threshold claim about total evening minutes in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C069: interaction claim about total daytime minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C070: interaction claim about total evening minutes and voice-mail plan in part ann_79: not supported by the model (the part's counterfactual difference has the same sign for only 89% of affected patients).
- C073: threshold claim about total night-time minutes in part ann_79: not supported by the model (the folds are spread over 39% of the observed range (allowed 20%), also within each category group: no single threshold; the effect past the fold is too small or not consistent across patients).
- C074: threshold claim about total evening minutes in part ann_79: not supported by the model (the effect past the fold is too small or not consistent across patients; no training patient is past the fold).
- C075: subgroup claim about state and voice-mail plan in part ann_79: not supported by the model (its effect goes both ways across the patients it acts on).
- C118: boundary claim about number of calls to customer service and total daytime minutes in part ann_83: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

## 1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (35):** `-11` (state:18), `-12` (state:19), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-2` (state:1), `-23` (state:29), `-24` (state:3), `-25` (state:30), `-28` (state:33), `-29` (state:34), `-31` (state:36), `-33` (state:38), `-35` (state:4), `-38` (state:42), `-39` (state:43), `-4` (state:11), `-40` (state:44), `-47` (state:50), `-48` (state:6), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-51` (state:9), `-53` (areaCode:408), `-54` (areaCode:415), `-6` (state:13), `-60` (totalDayCalls), `-63` (totalEveCalls), `-64` (totalEveCharge), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-7` (state:14), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

## 1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, ...) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)
  ▷ A2: linear_backbone (leaf, section 1.6.2)
  ▷ A3: vmail_state41_unit (leaf, section 1.6.3)
  ▷ A4: intl_plan_switch (leaf, section 1.6.4)
  ▷ A5: state_offsets (leaf, section 1.6.5)

▷ A6: usage_response (composition, section 1.6.6)
▷ A7: usage_hinge (leaf, section 1.6.7)

## 1.4 Network figures

## 1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is $1.22 \times 10^{-15}$ (tolerance $1.00 \times 10^{-9}$).

- #71: split `-37` into `-37_a`, `-37_b`, `-37_c`, `-37_d`.
- #72: split `-57` into `-57_a`, `-57_b`.
- #73: split `-62` into `-62_a`, `-62_b`.
- #74: split `-65` into `-65_a`, `-65_b`.
- #75: split `-41` into `-41_a`, `-41_b`.
- #76: identity node `identity_lin` before `0` (27 connection(s)).
- #77: identity node `identity_usage` before `0` (3 connection(s)).

## 1.6 Annotations

### 1.6.1 A1: churn_logit

**Kind:** Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6).

- **Entry nodes:** `-71` (numberCustomerServiceCalls), `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-64` (totalEveCharge), `-63` (totalEveCalls), `-61` (totalDayCharge), `-60` (totalDayCalls), `-59` (totalDayMinutes), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-43` (state:47), `-42` (state:46), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-36` (state:40), `-35` (state:4), `-34` (state:39), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-28` (state:33), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-23` (state:29), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-11` (state:18), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), … (18 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `582`→`0` (-1.4), `2067`→`0` (-0.262), `5667`→`0` (0.0263), `5821`→`0` (-0.0979), `6711`→`0` (1.68), `12726`→`0` (0.0229), `13164`→`0` (0.135), `13420`→`0` (-0.122), `identity_lin`→`0` (1), `identity_usage`→`0` (1).
- **With its descendants:** 95 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 94 candidate nodes.

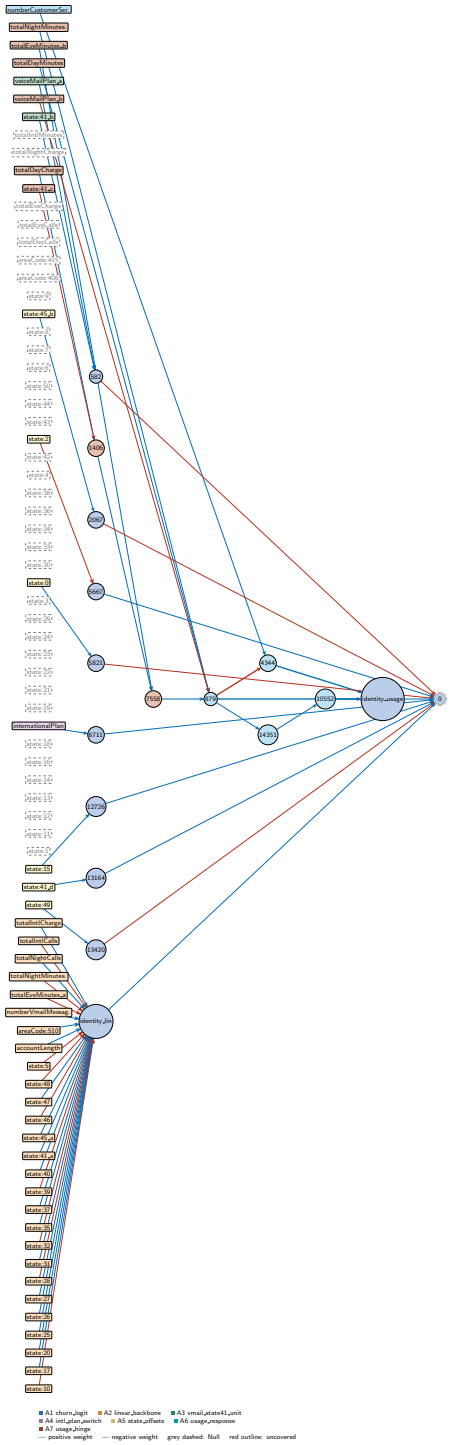



Figure 1: churn100-s7-c1: full view of the explained model state.

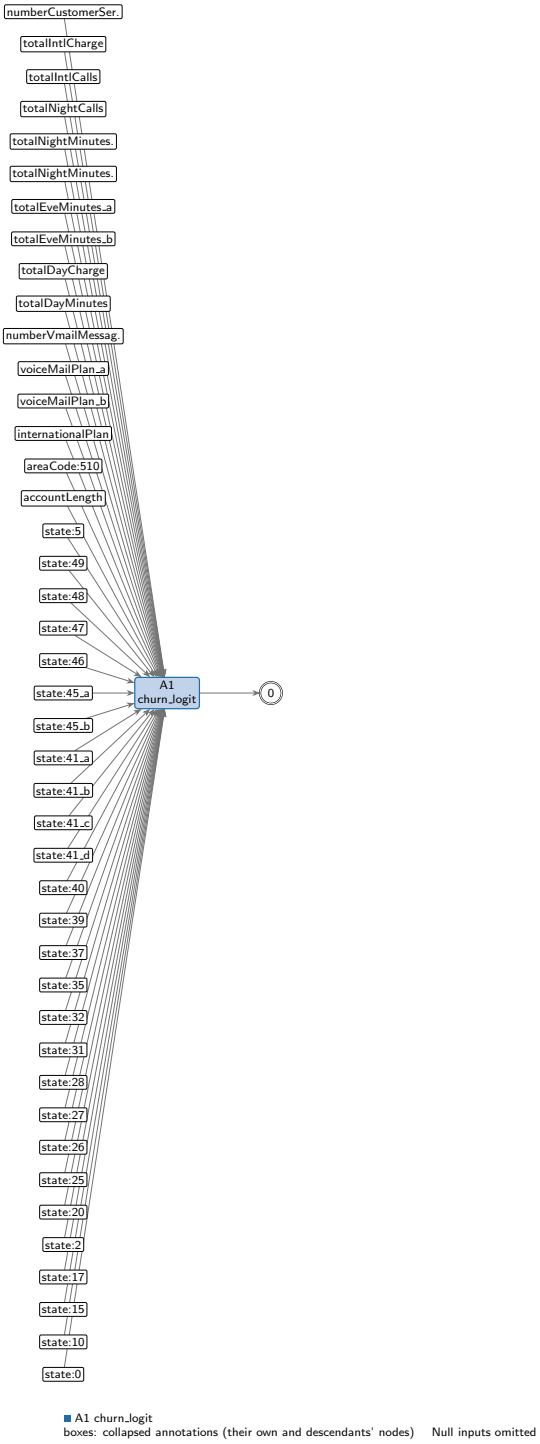



Figure 2: churn100-s7-c1: collapsed view at hierarchy level 0 (fully recomposed).

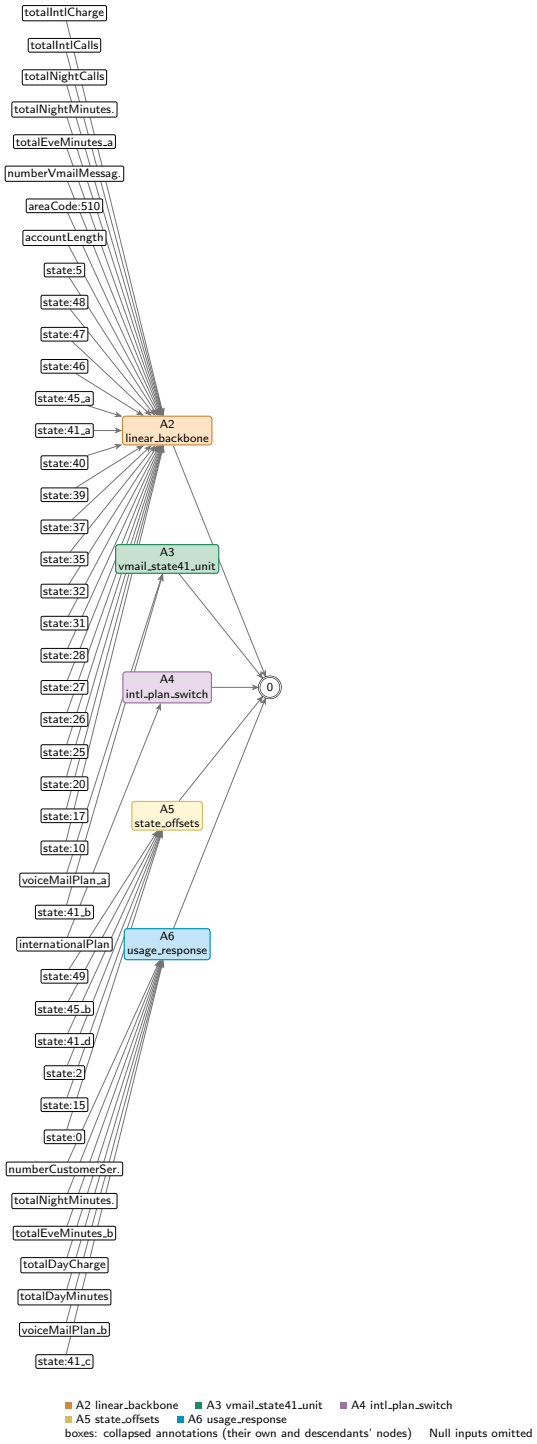



Figure 3: churn100-s7-c1: collapsed view at hierarchy level 1.

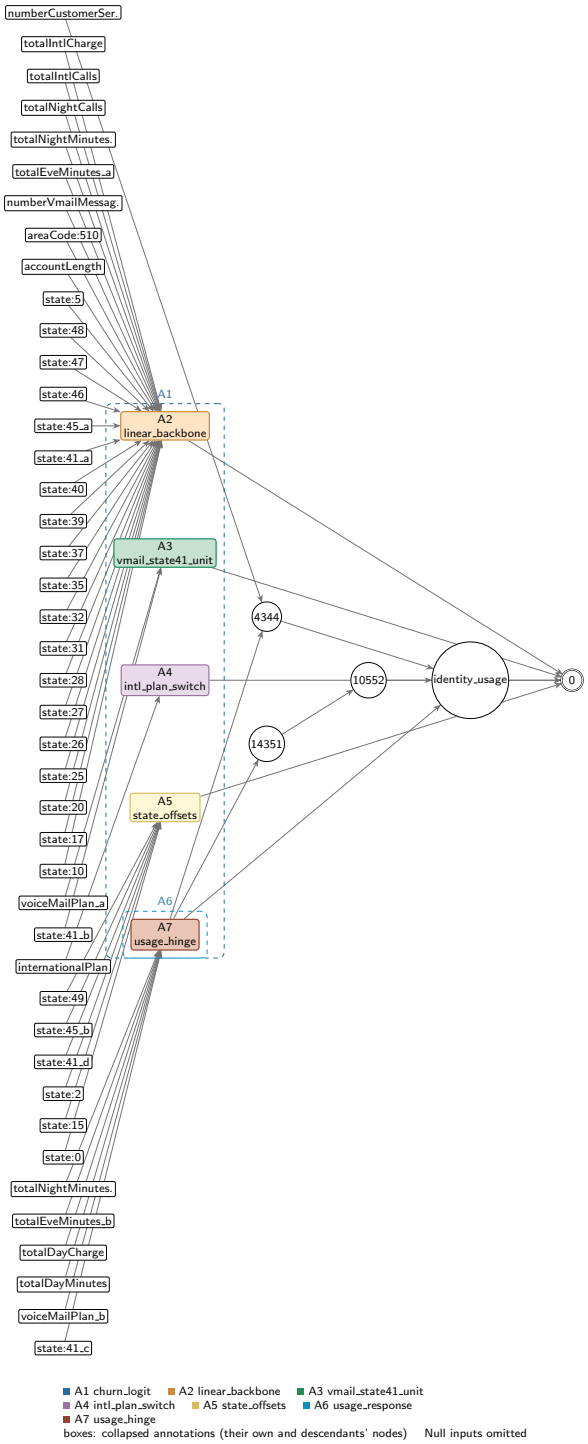



Figure 4: churn100-s7-c1: collapsed view at hierarchy level 2.

**Hypothesis.** Root: logit P(churn) = -1.96 + linear_backbone (identity_lin) + usage_response (identity_usage) + 1.68*6711 (intl_plan_switch) - 1.40*582 (vmail_state41_unit) + state_offsets (-0.26*2067 - 0.10*5821 + 0.02*12726 + 0.13*13164 - 0.12*13420 + 0.03*5667), purely additive in the logit; the sigmoid is the only coupling. 35 inputs are unconnected (international minutes, night charge, evening charge and calls, day calls, area codes 408/415 and 29 state indicators) and have zero effect. Train AUC 0.9189, accuracy 0.921. Performance sources (train): all units replaced by least-squares linear fits gives AUC 0.8336 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring usage_response alone gives 0.9189 (all the gain); its parts alone: usage hinge 0.8803, heavy-use hinge 0.8870, service-calls gate 0.8731. The international-plan, voicemail and state units are exact switches/offsets and restore nothing. Rare-category switches: state:46 -1.89 logit (80 rows, empirical log-odds offset -0.91); state:39 -1.23 (82, -0.93); state:47 +1.10 (72, +0.92); state:5 -1.09 (78, -0.51); state:45 -0.97 (91, -0.85); state:26 +0.81 (75, +0.50); state:49 -0.69 (122, -0.32); state:0 -0.68 (56, -0.76); state:10 -0.63 (60, -0.22); state:41 +0.25 mean, range -3.85..+0.43 (66, -0.04: it switches off the 1406 day-charge hinge, so heavy users lose up to 3.85).

**Closed form** (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0)\\
z_0 &= -1.95 + 1.0\,\text{linear_backbone}\big(x_{state:17}, x_{state:20},\\
&\quad x_{state:25}, x_{state:26}, x_{state:27}, x_{state:28}, x_{state:31}, x_{state:32},\\
&\quad x_{state:10}, x_{state:35}, x_{state:37}, x_{state:39}, x_{state:40}, x_{state:41_a},\\
&\quad x_{state:45_a}, x_{state:46}, x_{state:47}, x_{state:48}, x_{state:5}, x_{accountLength},\\
&\quad x_{areaCode:510}, x_{numberVmailMessages}, x_{totalEveMinutes_a}, x_{totalNightMinutes_a},\\
&\quad x_{totalNightCalls}, x_{totalIntlCalls}, x_{totalIntlCharge}\big)\\
&\quad + 1.0\,\text{usage_response}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b},\\
&\quad x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, x_{totalDayCharge},\\
&\quad x_{numberCustomerServiceCalls}\big)\\
&\quad + 0.0229\,\text{state_offsets_2}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)\\
&\quad + 0.0263\,\text{state_offsets_5}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)\\
&\quad + 0.135\,\text{state_offsets_3}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)\\
&\quad + 1.68\,\text{intl_plan_switch}\big(x_{internationalPlan}\big)\\
&\quad - 0.262\,\text{state_offsets_0}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)\\
&\quad - 1.4\,\text{vmail_state41_unit}\big(x_{voiceMailPlan_a}, x_{state:41_b}\big)\\
&\quad - 0.0979\,\text{state_offsets_1}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)\\
&\quad - 0.122\,\text{state_offsets_4}\big(x_{state:2}, x_{state:49}, x_{state:41_d},\\
&\quad x_{state:15}, x_{state:0}, x_{state:45_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma(z_0)\\
z_0 &= -1.95 + 3.06\,\text{relu}\big(-0.638 + 1.17\,\text{relu}\big(-0.158\\
&\quad + 1.08\,\text{relu}\big(-0.993 + 0.219x_{totalNightMinutes_b}\\
&\quad + 0.527x_{totalDayMinutes} + 0.554\,\text{relu}\big(0.515\\
&\quad + 1.21\,\text{relu}\big(-0.503 + 0.545x_{totalDayCharge}\\
&\quad - 0.885x_{state:41_c}\big) + 0.665x_{totalEveMinutes_b}\big)\\
&\quad - 1.01x_{voiceMailPlan_b}\big)\big)\big) + 0.321x_{totalIntlCharge}
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.0229\,\mathrm{relu}\big(0.47 + 0.878x_{state:15}\big) \\
&+ 0.199x_{numberVmailMessages} + 2.57\,\mathrm{relu}\big(-0.993 \\
&+ 0.219x_{totalNightMinutes_b} + 0.527x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545x_{totalDayCharge} - 0.885x_{state:41_c}\big) + 0.665x_{totalEveMinutes_b}\big) \\
&- 1.01x_{voiceMailPlan_b}\big) + 0.042x_{state:35} + 0.0442x_{areaCode:510} \\
&+ 0.0249x_{totalNightCalls} + 0.0263\,\mathrm{relu}\big(-0.325 \\
&- 0.0317x_{state:2}\big) + 0.0544x_{state:41_a} + 0.0545x_{state:40} \\
&+ 0.0145x_{state:25} + 0.00187x_{state:20} + 0.0301x_{state:17} \\
&+ 0.135\,\mathrm{relu}\big(-0.236 + 1.02x_{state:41_d}\big) + 0.146x_{state:47} \\
&+ 0.0375x_{state:27} + 0.0379x_{state:37} + 0.0871x_{accountLength} \\
&+ 0.0223x_{state:45_a} + 2.88\,\mathrm{relu}\big(-0.384 \\
&+ 0.748x_{numberCustomerServiceCalls} - 3.24\,\mathrm{relu}\big(-0.993 \\
&+ 0.219x_{totalNightMinutes_b} + 0.527x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545x_{totalDayCharge} - 0.885x_{state:41_c}\big) + 0.665x_{totalEveMinutes_b}\big) \\
&- 1.01x_{voiceMailPlan_b}\big)\big) + 0.049x_{state:31} \\
&+ 1.68\,\mathrm{relu}\big(-0.228 + 0.635x_{internationalPlan}\big) \\
&+ 0.109x_{state:26} - 0.0289x_{state:28} - 0.139x_{totalEveMinutes_a} - 0.175x_{state:39} \\
&- 0.262\,\mathrm{relu}\big(-0.691 + 0.757x_{state:45_b}\big) \\
&- 0.264x_{totalIntlCalls} - 0.0184x_{state:32} - 0.0761x_{state:10} \\
&- 0.114x_{totalNightMinutes_a} - 0.0165x_{state:48} - 0.264x_{state:46} \\
&- 0.151x_{state:5} - 1.4\,\mathrm{relu}\big(1.51 + 0.17x_{voiceMailPlan_a} \\
&+ 0.0942x_{state:41_b}\big) - 0.0979\,\mathrm{relu}\big(-1.7 + 1.02x_{state:0}\big) \\
&- 0.122\,\mathrm{relu}\big(0.00575 + 1.01x_{state:49}\big)
\end{aligned}
$$

where $y_0$ is node `0`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00775, and the exact function by at most $7.77 \times 10^{-16}$.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.96 + linear_backbone (identity_lin) + usage_response (identity_usage) + 1.68*6711 (intl_plan_switch) - 1.40*582 (vmail_state41_unit) + state_offsets (-0.26*2067 - 0.10*5821 + 0.02*12726 + 0.13*13164 - 0.12*13420 + 0.03*5667), purely additive in the logit; the sigmoid is the only coupling. 35 inputs are unconnected (international minutes, night charge, evening charge and c

**Authorship:** drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

### 1.6.2 A2: linear_backbone

**Kind:** Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-10` (state:17), `-14` (state:20), `-19` (state:25), `-20` (state:26), `-21` (state:27), `-22` (state:28), `-26` (state:31), `-27` (state:32), `-3` (state:10), `-30` (state:35), `-32` (state:37), `-34` (state:39), `-36` (state:40), `-37_a` (state:41_a), `-41_a` (state:45_a), `-42` (state:46), `-43` (state:47), `-44` (state:48), `-46` (state:5), `-52` (accountLength), `-55` (areaCode:510), `-58` (numberVmailMessages), `-62_a` (totalEveMinutes_a), `-65_a` (totalNightMinutes_a), `-66` (totalNightCalls), `-69` (totalIntlCalls), `-70` (totalIntlCharge).

- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-3`→`identity_lin` (-0.0761), `-10`→`identity_lin` (0.0301), `-14`→`identity_lin` (0.00187), `-19`→`identity_lin` (0.0145), `-20`→`identity_lin` (0.109), `-21`→`identity_lin` (0.0375), `-22`→`identity_lin` (-0.0289), `-26`→`identity_lin` (0.049), `-27`→`identity_lin` (-0.0184), `-30`→`identity_lin` (0.042), `-32`→`identity_lin` (0.0379), `-34`→`identity_lin` (-0.175), `-36`→`identity_lin` (0.0545), `-42`→`identity_lin` (-0.264), `-43`→`identity_lin` (0.146), `-44`→`identity_lin` (-0.0165), `-46`→`identity_lin` (-0.151), `-52`→`identity_lin` (0.0871), `-55`→`identity_lin` (0.0442), `-58`→`identity_lin` (0.199), `-66`→`identity_lin` (0.0249), `-69`→`identity_lin` (-0.264), `-70`→`identity_lin` (0.321), `-37_a`→`identity_lin` (0.0544), `-62_a`→`identity_lin` (-0.139), `-65_a`→`identity_lin` (-0.114), `-41_a`→`identity_lin` (0.0223).
- **Covers** (coverage, own and as part of the hierarchy): 27 candidate nodes.

**Hypothesis.** Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 27 inputs, all with small weights (at most 0.32 per SD): international charge +0.32, international calls -0.26, voicemail messages +0.20, evening minutes -0.14 and night minutes -0.11 (these two also feed the usage hinge with positive weights), account length +0.09, area code 510 +0.04 (+0.10 logit), and state offsets: state:46 -1.89 logit (80 customers), state:39 -1.23 (82), state:47 +1.10 (72), state:5 -1.09 (78), state:26 +0.81, state:10 -0.63, state:40 +0.43, state:41 +0.43, state:31 +0.33, then smaller. International plan, voicemail plan, day minutes and customer-service calls have NO direct edge. Evidence: closed form exact (identity, R2 1.0); train sensitivities equal the weights (international charge 0.32, international calls 0.26, state:46 0.26, voicemail messages 0.20). Mean ablation AUC 0.9076 (full 0.9189).

**Closed form:**

$$
\begin{aligned}
y_0 = {} & 0.321x_{totalIntlCharge} + 0.199x_{numberVmailMessages} + 0.042x_{state:35} \\
& + 0.0442x_{areaCode:510} + 0.0249x_{totalNightCalls} + 0.0544x_{state:41_a} \\
& + 0.0545x_{state:40} + 0.0145x_{state:25} + 0.00187x_{state:20} + 0.0301x_{state:17} \\
& + 0.146x_{state:47} + 0.0375x_{state:27} + 0.0379x_{state:37} + 0.0871x_{accountLength} \\
& + 0.0223x_{state:45_a} + 0.049x_{state:31} + 0.109x_{state:26} - 0.0289x_{state:28} \\
& - 0.139x_{totalEveMinutes_a} - 0.175x_{state:39} - 0.264x_{totalIntlCalls} \\
& - 0.0184x_{state:32} - 0.0761x_{state:10} - 0.114x_{totalNightMinutes_a} \\
& - 0.0165x_{state:48} - 0.264x_{state:46} - 0.151x_{state:5}
\end{aligned}
$$

where $y_0$ is node `identity_lin`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00352, and the exact function by at most $8.88 \times 10^{-16}$.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 27 inputs, all with small weights (at most 0.32 per SD): international charge +0.32, international calls -0.26, voicemail messages +0.20, evening minutes -0.14 and night minutes -0.11 (these two also feed the usage hinge with positive weights), account length +0.09, area code 510 +0.04 (+0.10 logit),

**Authorship:** drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

### 1.6.3 A3: vmail_state41_unit

**Kind:** Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-57_a` (voiceMailPlan_a), `-37_b` (state:41_b).
- **Exit nodes:** `582`.
- **Internal nodes:** none.
- **Edges (weight):** `-37_b`→`582` (0.0942), `-57_a`→`582` (0.17).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

**Hypothesis.** 582 = relu(1.51 + 0.17 vmailPlan + 0.09 state:41), logit weight -1.40. The pre-activation stays between 1.40 and 2.52 on train, so the unit is always active and exactly affine (R2 1.0): a constant -1.96 logit for customers without a voicemail plan outside state:41, -0.54 extra for a voicemail plan and -1.04 extra for state:41. Hypothesis: a fixed offset plus a voicemail-plan and state:41 discount. Evidence: sensitivity 0.17 (voicemail plan), 0.09 (state:41); linear replacement AUC 0.9189 (unchanged), mean ablation 0.9184.

**Closed form:**

$$y_0 = \mathrm{relu}\big(1.51 + 0.17 x_{voiceMailPlan_a} + 0.0942 x_{state:41_b}\big)$$

where $y_0$ is node `582`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00187, and the exact function by at most 0.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 582 = relu(1.51 + 0.17 vmailPlan + 0.09 state:41), logit weight -1.40. The pre-activation stays between 1.40 and 2.52 on train, so the unit is always active and exactly affine (R2 1.0): a constant -1.96 logit for customers without a voicemail plan outside state:41, -0.54 extra for a voicemail plan and -1.04 extra for state:41. Hypothesis: a fixed offset plus a voicemail-plan and state:41 discount.

**Authorship:** drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

### 1.6.4 A4: intl_plan_switch

**Kind:** Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-56` (internationalPlan).
- **Exit nodes:** `6711`.
- **Internal nodes:** none.
- **Edges (weight):** `-56`→`6711` (0.635).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

**Hypothesis.** 6711 = relu(-0.23 + 0.64 intlPlan), logit weight +1.68: exactly 0 for customers without an international plan and +2.91 logit for the 380 with one (R2 1.0). Hypothesis: an international-plan switch, the model's only route for the plan (no direct edge). Empirical churn 0.42 with the plan vs 0.11 without; model 0.45 vs 0.11. No interaction with international usage (international minutes are unconnected; international charge and calls act linearly in the backbone).

Evidence: line plot correlation 1.00; mean ablation AUC 0.8611; linear replacement 0.9189 (identical, exact switch).

**Closed form:**

$$y_0 = \text{relu}\big(-0.228 + 0.635 x_{internationalPlan}\big)$$

where $y_0$ is node `6711`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00131, and the exact function by at most 0.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 6711 = relu(-0.23 + 0.64 intlPlan), logit weight +1.68: exactly 0 for customers without an international plan and +2.91 logit for the 380 with one (R2 1.0). Hypothesis: an international-plan switch, the model's only route for the plan (no direct edge). Empirical churn 0.42 with the plan vs 0.11 without; model 0.45 vs 0.11. No interaction with international usage (international minutes are unconnec

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#### 1.6.5 A5: state_offsets

**Kind:** Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-13` (state:2), `-45` (state:49), `-37_d` (state:41_d), `-8` (state:15), `-1` (state:0), `-41_b` (state:45_b).
- **Exit nodes:** `2067`, `5821`, `12726`, `13164`, `13420`, `5667`.
- **Internal nodes:** none.
- **Edges (weight):** `-1`→`5821` (1.02), `-13`→`5667` (-0.0317), `-8`→`12726` (0.878), `-45`→`13420` (1.01), `-37_d`→`13164` (1.02), `-41_b`→`2067` (0.757).
- **Covers** (coverage, own and as part of the hierarchy): 6 candidate nodes.

**Hypothesis.** Six single-state units, each a function of one binary indicator and so exactly an offset (R2 1.0): 2067 state:45 -1.12 logit (91 customers; with its direct +0.15, net -0.97), 13164 state:41 +1.04 (66; it offsets most of state:41's -1.04 in vmail_state41_unit), 13420 state:49 -0.69 (122), 5821 state:0 -0.68 (56), 12726 state:15 +0.14 (82; always active, constant +0.008), 5667 state:2 is never active (dead unit, zero effect). Hypothesis: per-state offsets. Evidence: sensitivities as listed; linear replacement AUC 0.9189 (unchanged); mean ablation 0.9167.

**Closed form:**

$$\begin{aligned}
y_0 &= \text{relu}\big(-0.691 + 0.757 x_{state:45_b}\big)\\
y_1 &= \text{relu}\big(-1.7 + 1.02 x_{state:0}\big)\\
y_2 &= \text{relu}\big(0.47 + 0.878 x_{state:15}\big)\\
y_3 &= \text{relu}\big(-0.236 + 1.02 x_{state:41_d}\big)\\
y_4 &= \text{relu}\big(0.00575 + 1.01 x_{state:49}\big)\\
y_5 &= \text{relu}\big(-0.325 - 0.0317 x_{state:2}\big)
\end{aligned}$$

where $y_0$ is node `2067`; $y_1$ is node `5821`; $y_2$ is node `12726`; $y_3$ is node `13164`; $y_4$ is node `13420`; $y_5$ is node `5667`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0381, and the exact function by at most 0.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Six single-state units, each a function of one binary indicator and so exactly an offset (R2 1.0): 2067 state:45 -1.12 logit (91 customers; with its direct +0.15, net -0.97), 13164 state:41 +1.04 (66; it offsets most of state:41's -1.04 in vmail_state41_unit), 13420 state:49 -0.69 (122), 5821 state:0 -0.68 (56), 12726 state:15 +0.14 (82; always active, constant +0.008), 5667 state:2 is never activ

**Authorship:** drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

#### 1.6.6 A6: usage_response

**Kind:** Composition annotation (depth 1); parent A1 (section 1.6.1); children A7 (section 1.6.7).

- **Entry nodes:** `-65_b` (totalNightMinutes_b), `-57_b` (voiceMailPlan_b), `-59` (totalDayMinutes), `-62_b` (totalEveMinutes_b), `-37_c` (state:41_c), `-61` (totalDayCharge), `-71` (numberCustomerServiceCalls).
- **Exit nodes:** `identity_usage`.
- **Internal nodes:** `4344`, `14351`, `10552`.
- **Edges (weight):** `-71`→`4344` (0.748), `879`→`4344` (-3.24), `879`→`14351` (1.08), `14351`→`10552` (1.17), `879`→`identity_usage` (2.57), `4344`→`identity_usage` (2.88), `10552`→`identity_usage` (3.06).
- **With its descendants:** 14 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 13 candidate nodes.

**Hypothesis.** Composition on usage_hinge: identity_usage = 2.57*879 + 3.06*10552 + 2.88*4344 enters the logit with weight 1. 10552 = relu(-0.64 + 1.17*relu(-0.16 + 1.08*879)) is a second, steeper hinge that opens once 879 exceeds 0.65 (about 240 day minutes at median other use; active for 12% of customers), so the usage effect accelerates. 4344 = relu(-0.38 + 0.75 calls - 3.24*879) is a service-calls gate: shut for 0-2 calls, open from 3 calls, and closed again by usage (at 3 calls once 879 > 0.14, at 5 calls once 879 > 0.49). Hypothesis: usage and service calls interact non-additively: service calls raise churn only for low-usage customers (or voicemail-plan customers, for whom usage is off), heavy users get the usage hinges instead. Evidence (train): usage logit (879+10552) by day minutes 0.14 (<200), 0.58 (200-220), 1.08 (220-240), 2.02 (240-260), 3.43 (260-280), 6.37 (>=280); churn 0.10/0.06/0.14/0.26/0.46/0.67 vs model 0.09/0.09/0.17/0.30/0.49/0.69. Without a voicemail plan churn reaches 0.84 above 280 minutes (model 0.86), with a plan 0.04 (model 0.09). Gate logit at 0-7 calls: 0/0/0/0.80/2.11/3.59/5.30/7.08 (1.25/2.90/4.55/6.20 at 3-6 calls when usage is off, 0.10/0.50/1.05/2.01 when it is on). Churn by calls 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 vs model 0.116/0.111/0.109/0.150/0.319/0.598/0.849/0.940: the gate opens one call too early (3 calls: +0.80 logit, empirical churn unchanged) and is over-confident from 6 calls (39 customers). Mean ablation of all three terms AUC 0.6557 (the largest drop: full 0.9189); gate alone 0.8202, heavy-use hinge alone 0.9093; linear replacement 0.8336 (R2 0.53/0.29/0.53); restoring usage_response alone onto the all-linear model recovers 0.9189 (all-linear 0.8336), i.e. all of the gain.

**Closed form** (children as named functions):

$$
\begin{aligned}
y_0 = 3.06\,\mathrm{relu}\big(&-0.638 + 1.17\,\mathrm{relu}\big(-0.158 \\
&+ 1.08\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, \\
&x_{totalDayCharge}\big)\big)\big) \\
&+ 2.57\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state:41_c}, x_{totalDayCharge}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 2.88\,\mathrm{relu}\big(-0.384 + 0.748 x_{numberCustomerServiceCalls} \\
&- 3.24\,\mathrm{usage_hinge}\big(x_{totalNightMinutes_b}, x_{voiceMailPlan_b}, \\
&x_{totalDayMinutes}, x_{totalEveMinutes_b}, x_{state{:}41_c}, x_{totalDayCharge}\big)\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = 3.06\,\mathrm{relu}\big(&-0.638 + 1.17\,\mathrm{relu}\big(-0.158 \\
&+ 1.08\,\mathrm{relu}\big(-0.993 + 0.219 x_{totalNightMinutes_b} \\
&+ 0.527 x_{totalDayMinutes} + 0.554\,\mathrm{relu}\big(0.515 \\
&+ 1.21\,\mathrm{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&- 0.885 x_{state{:}41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)\big)\big) + 2.57\,\mathrm{relu}\big(-0.993 \\
&+ 0.219 x_{totalNightMinutes_b} + 0.527 x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545 x_{totalDayCharge} - 0.885 x_{state{:}41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big) + 2.88\,\mathrm{relu}\big(-0.384 \\
&+ 0.748 x_{numberCustomerServiceCalls} - 3.24\,\mathrm{relu}\big(-0.993 \\
&+ 0.219 x_{totalNightMinutes_b} + 0.527 x_{totalDayMinutes} \\
&+ 0.554\,\mathrm{relu}\big(0.515 + 1.21\,\mathrm{relu}\big(-0.503 \\
&+ 0.545 x_{totalDayCharge} - 0.885 x_{state{:}41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)\big)
\end{aligned}
$$

where $y_0$ is node `identity_usage`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0254, and the exact function by at most $4.44 \times 10^{-15}$.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition on usage_hinge: identity_usage = 2.57*879 + 3.06*10552 + 2.88*4344 enters the logit with weight 1. 10552 = relu(-0.64 + 1.17*relu(-0.16 + 1.08*879)) is a second, steeper hinge that opens once 879 exceeds 0.65 (about 240 day minutes at median other use; active for 12% of customers), so the usage effect accelerates. 4344 = relu(-0.38 + 0.75 calls - 3.24*879) is a service-calls gate: shut

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### 1.6.7 A7: usage_hinge

**Kind:** Leaf annotation (depth 2); parent A6 (section 1.6.6).

- **Entry nodes:** `-65_b` (totalNightMinutes_b), `-57_b` (voiceMailPlan_b), `-59` (totalDayMinutes), `-62_b` (totalEveMinutes_b), `-37_c` (state:41_c), `-61` (totalDayCharge).
- **Exit nodes:** `879`.
- **Internal nodes:** `1406`, `7558`.
- **Edges (weight):** `-59`→`879` (0.527), `-61`→`1406` (0.545), `7558`→`879` (0.554), `1406`→`7558` (1.21), `-37_c`→`1406` (-0.885), `-57_b`→`879` (-1.01), `-62_b`→`7558` (0.665), `-65_b`→`879` (0.219).
- **Covers** (coverage, own and as part of the hierarchy): 8 candidate nodes.

**Hypothesis.** 879 = relu(-0.99 + 0.53 dayMin + 0.22 nightMin - 1.01 vmailPlan + 0.55*7558), 7558 = relu(0.51 + 0.67 eveMin + 1.21*1406), 1406 = relu(-0.50 + 0.54 dayCharge - 0.88 state:41). Day minutes and day charge are collinear (r = 1.0000), so 879 is a hinge in total usage, dominated by day use. Hypothesis: a usage hinge that is never active for voicemail-plan customers (0 of 1061) and active for 47% of the rest; at median evening/night use it switches on at about 190 day minutes and rises steadily, with a second slope from the 1406 day-charge hinge (on for 24% of customers, never in state:41). Its exit feeds the logit directly (weight 2.57) and drives the heavy-use hinge and the service-calls gate in usage_response. Evidence (train, n=4000): sensitivity of 879 to voicemail plan 0.35, day minutes 0.18, evening minutes 0.12, state:41 0.09, night minutes 0.08, day charge 0.05. Mean ablation of its direct logit term AUC 0.9017; linear replacement 0.9107 (R2 0.53); restored alone onto the all-linearised model 0.8803 (all-linear 0.8336).

**Closed form:**

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-0.993 + 0.219 x_{totalNightMinutes_b} \\
&+ 0.527 x_{totalDayMinutes} + 0.554 \operatorname{relu}\big(0.515 \\
&+ 1.21 \operatorname{relu}\big(-0.503 + 0.545 x_{totalDayCharge} \\
&- 0.885 x_{state:41_c}\big) + 0.665 x_{totalEveMinutes_b}\big) \\
&- 1.01 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

where $y_0$ is node `879`; $z_i$, where shown, is the pre-activation of a sigmoid output $y_i$; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00297, and the exact function by at most $4.44 \times 10^{-16}$.

**Evidence** (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 879 = relu(-0.99 + 0.53 dayMin + 0.22 nightMin - 1.01 vmailPlan + 0.55*7558), 7558 = relu(0.51 + 0.67 eveMin + 1.21*1406), 1406 = relu(-0.50 + 0.54 dayCharge - 0.88 state:41). Day minutes and day charge are collinear (r = 1.0000), so 879 is a hinge in total usage, dominated by day use. Hypothesis: a usage hinge that is never active for voicemail-plan customers (0 of 1061) and active for 47% of the

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### 1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft